3D Pedestrian Detection Method, System, Equipment, and Medium for Open Scenes

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Abstract

The present invention discloses a 3D pedestrian detection method, system, equipment, and medium for open scenes, relating to the field of pedestrian detection technology. It includes: encoding point cloud data into pillar features, introducing an external attention mechanism and an attention mechanism based on squeeze-and-excitation to supervise point channels, feature channels, and pillar channels, suppressing noise during the encoding process to obtain encoded pillar features; enhancing pedestrian scale perception capability through a spatial attention mechanism, and utilizing top-down and bottom-up bidirectional propagation paths for scale fusion, thereby performing 3D target pedestrian detection, enhancing the scale perception capability of pillar features, alleviating scale confusion problems in pedestrian detection, and simultaneously improving detection speed.

Claims

1. A 3D pedestrian detection method for open scenes, characterized in that it comprises:

   Acquiring point cloud data of a target pedestrian to be detected, encoding the point cloud data into pillar features, wherein the dimensions of the pillar features include point channels, feature channels, and pillar channels;

   Introducing an external attention mechanism to calculate attention weights for the point channels and the feature channels respectively, obtaining a first weight map based on the attention weight map of the point channels and the attention weight map of the feature channels, concatenating the first weight map with the center point on the pillar channels, and then squeezing the point channels and feature channels to obtain a second weight map, weighting the second weight map according to the first weight map, and concatenating it with the pillar features to obtain encoded pillar features;

   Projecting the encoded pillar features to obtain a pseudo-image, extracting multi-scale features from the pseudo-image, generating a spatial attention weight map for each scale feature, and performing element-wise multiplication and element-wise addition with the corresponding scale features to obtain pseudo-image features with scale perception;

   For the pseudo-image features at multiple scales, performing scale fusion using top-down and bottom-up bidirectional propagation paths, and thereby performing 3D target pedestrian detection based on the fused feature maps.

2. The 3D pedestrian detection method for open scenes according to claim 1, characterized in that the process of obtaining the first weight map comprises: performing max-pooling operations on the point channels and feature channels respectively, introducing two different memory units M_k and M_v as key and value; performing element-wise multiplication on the max-pooled point channels and feature channels with M_k, and then performing element-wise multiplication with M_v, thereby obtaining the attention weight map of the point channels and the attention weight map of the feature channels, and multiplying the attention weight map of the point channels and the attention weight map of the feature channels to obtain the first weight map.

3. The 3D pedestrian detection method for open scenes according to claim 1, characterized in that the process of generating the spatial attention weight map comprises: performing average-pooling and max-pooling operations on each scale feature, concatenating the results of average-pooling and max-pooling, and then using a convolution kernel to perform an excitation operation, thereby generating the spatial attention weight map.

4. The 3D pedestrian detection method for open scenes according to claim 1, characterized in that it comprises performing element-wise multiplication of the spatial attention weight map of each scale with the feature under the corresponding scale, and then performing element-wise addition with the feature under the corresponding scale to obtain pseudo-image features with scale perception.

5. The 3D pedestrian detection method for open scenes according to claim 1, characterized in that the process of performing scale fusion using top-down and bottom-up bidirectional propagation paths includes a top-down aggregation path network and a bottom-up aggregation path network; Specifically, the obtained pseudo-image features at multiple scales are P_i^{in}, where i denotes the i-th scale, $i = 1, 2, 3$. According to the second-level pseudo-image feature and the third-level pseudo-image feature, obtaining the aggregation path network for top-down propagation P_2^{up}, where P_2^{up} assigns weights and fuses the third-level pseudo-image feature and the second-level pseudo-image feature; According to the first-level pseudo-image feature P_1^{in} and obtaining the first propagation layer P_1^{out} in the bottom-up aggregation path network P_i^{out}, where P_1^{out} assigns weights and fuses P_1^{in} and P_2^{in}; subsequently, based on the first propagation layer, obtaining the second propagation layer P_2^{out} according to the second-level pseudo-image feature and P_2^{up}, where P_2^{out} assigns weights and fuses P_1^{out}, P_2^{in}, and P_2^{up}; finally, based on the second propagation layer, obtaining the third propagation layer P_3^{out} according to the third-level pseudo-image feature, where P_3^{out} assigns weights and fuses P_2^{out} and P_3^{in}.

6. The 3D pedestrian detection method for open scenes according to claim 5, characterized in that the top-down aggregation path network is:

$$P_2^{up} = Conv\left(\frac{w_1 \cdot P_2^{in} + w_2 \cdot Resize(P_3^{in})}{w_1 + w_2 + \epsilon}\right)$$

Where $Resize$ is an upsampling or downsampling operation; $Conv$ is a convolution operation; ϵ is a constant; w_1, w_2 respectively represent the weights of relatively shallow features and relatively deep features in the fusion path.

7. The 3D pedestrian detection method for open scenes according to claim 5, characterized in that the bottom-up aggregation path network is:

$$P_1^{out} = Conv\left(\frac{w_1 \cdot P_2^{in} + w_2 \cdot Resize(P_3^{in})}{w_1 + w_2 + \epsilon}\right)$$

$$P_2^{out} = Conv\left(\frac{w_1'' \cdot P_2^{in} + w_2'' \cdot P_2^{up} + w_3'' \cdot Resize(P_1^{out})}{w_1'' + w_2'' + w_3'' + \epsilon}\right)$$

$$P_3^{out} = Conv\left(\frac{w_1''' \cdot P_3^{in} + w_2''' \cdot Resize(P_2^{out})}{w_1''' + w_2''' + \epsilon}\right)$$

Where P_i^{out} is the bottom-up aggregation path network, $i = 1, 2, 3$; $Resize$ is an upsampling or downsampling operation; $Conv$ is a convolution operation; ϵ is a constant; $w_1, w_1', w_1'', w_1''', w_2''$ respectively represent the weights of relatively shallow features in the fusion path; w_2, w_2', w_2''', w_3'' respectively represent the weights of relatively deep features in the fusion path.

8. A 3D pedestrian detection system for open scenes, characterized in that it comprises: A point cloud processing module configured to acquire point cloud data of a target pedestrian to be detected, encode the point cloud data into pillar features, where dimensions of the pillar features include point channels, feature channels, and pillar channels; A multi-attention processing module configured to introduce an external attention mechanism to respectively calculate attention weights for point channels and feature channels, obtain a first weight map based on the attention weight map of the point channels and the attention weight map of the feature channels, concatenate the first weight map with the center point on the pillar channels, and then squeeze the point channels and feature channels to obtain a second weight map, weight the second weight map according to the first weight map, and concatenate it with the pillar features to obtain encoded pillar features; A scale perception module configured to project the encoded pillar features to obtain a pseudo-image, extract multi-scale features from the pseudo-image, generate a spatial attention weight map for each scale feature, and perform element-wise multiplication and element-wise addition with the corresponding scale features to obtain pseudo-image features with scale perception; A scale fusion module configured to fuse the multi-scale pseudo-image features using top-down and bottom-up bidirectional propagation paths, and thereby perform 3D target pedestrian detection based on the fused feature maps.

9. An electronic device, characterized in that it comprises a memory and a processor, and computer instructions stored on the memory and executed on the processor, which, when executed by the processor, perform the method according to any one of claims 1 to 7.

10. A computer-readable storage medium, characterized in that it is used to store computer instructions, which, when executed by a processor, perform the method according to any one of claims 1 to 7.

Description

Technical Field

The present invention relates to the technical field of pedestrian detection, and in particular, to a 3D pedestrian detection method, system, equipment, and medium for open scenes.

Background Art

Currently, methods for real-time 3D target pedestrian detection based on deep learning mainly use grid-based point cloud representations, which are roughly divided into 3D voxel-based and pillar-based methods. Voxel-based methods typically segment input point clouds into regular 3D voxel grids, and then use 3D convolution encoders to learn multi-level geometric representations... However, even with 3D sparse convolutions, it is difficult to achieve both high detection accuracy and fast detection speed simultaneously. Pillar-based methods project 3D point clouds

onto a 2D pseudo-image plane, and then fuse multi-scale features on a feature pyramid network... However, using a simple fully connected layer for encoding is insufficient, and the above methods ignore the importance of encoding pillar features, which directly affects the quality of 2D pseudo-image generation.

Specific Embodiments

The present invention proposes a multi-attention module to encode pillar features, improving their expressive power.

1. Multi-Attention Module

The multi-attention module calculates the first weight map F^{out} using memory units M_k and M_v:

$$F_p^{out} = A_p M_v, \quad A_p = Norm(F_p^{max} M_k^T)$$

$$F_c^{out} = A_c M_v, \quad A_c = Norm(F_c^{max} M_k^T)$$

$$F_1^{out} = F_c^{out} \times F_p^{out}$$

where A_c is the attention weight map of the feature channels, A_p is the attention weight map of the point channels, M_k and M_v are learnable parameters independent of the input, $Norm$ is a normalization function, and T represents the transpose operation.

Further, the second weight map A_{pi} and final encoded features F_2^{out} are:

$$A_{pi} = FC(FC([P_c; F_1^{out}]))$$

$$F_2^{out} = FC(F) + \sigma(A_{pi}) \times F_1^{out}$$

where FC is a fully connected layer, P_c is the center point, σ is the sigmoid activation function, and $[\cdot;\cdot]$ denotes concatenation.

2. Spatial Attention Weight Map

The spatial attention weight map is denoted as:

$$M_s(P_i) = \sigma(Conv([AvgPool(P_i); MaxPool(P_i)])) = \sigma(Conv([P_{avg}; P_{max}]))$$

3. Loss Functions

The overall loss function $\mathcal{L}_{total}$ is designed as:

$$\mathcal{L}_{total} = \frac{1}{N_{pos}}(\beta_{res}\mathcal{L}_{res} + \beta_{cls}\mathcal{L}_{cls} + \beta_{dir}\mathcal{L}_{dir})$$

The 3D bounding box regression loss is represented as:

$$\Delta x = \frac{x^{gt} - x^p}{d^p}, \quad \Delta y = \frac{y^{gt} - y^p}{d^p}, \quad \Delta z = \frac{z^{gt} - z^p}{h^p}$$

$$\Delta w = \log\frac{w^{gt}}{w^p}, \quad \Delta l = \log\frac{l^{gt}}{l^p}, \quad \Delta h = \log\frac{h^{gt}}{h^p}$$

$$\Delta\theta = \sin(\theta^{gt} - \theta^p)$$

$$\mathcal{L}_{loc} = \sum_{b\in(x,y,z,w,l,h,\theta)} SmoothL1(\Delta b)$$

The classification loss $\mathcal{L}_{cls}$ is:

$$\mathcal{L}_{cls} = -c(1-q_t)^{\gamma}\log q_t$$

Experimental Results

Table 1: Quantitative analysis experimental results

Method	Stage	Avg. Precision in 3D (IoU 0.5)				Avg. Precision in 2D BEV (IoU 0.5)				FPS
		Easy	Mod	Hard	Avg	Easy	Mod	Hard	Avg	
PointPainting	Two-stage	50.32	40.97	37.87	43.05	58.70	49.93	46.26	51.64	3
FromVoxelToPoint	Two-stage	51.80	43.28	45.26	40.71	56.54	48.15	45.63	50.11	10
F-PointPillars	One-stage	51.22	42.82	39.28	44.46	60.98	52.23	48.30	53.84	14
HMFI	Two-stage	50.88	42.65	39.78	44.44	55.61	47.77	45.17	49.51	9
EPNet++	Two-stage	52.79	44.38	46.15	41.29	56.54	48.47	44.55	49.85	10
Second	One-stage	51.07	42.56	37.29	43.64	55.10	46.27	44.76	48.71	20
PointPillars	One-stage	51.45	41.92	44.09	38.89	57.60	48.64	45.78	50.67	63
TANet	One-stage	53.72	44.34	40.49	46.18	60.85	51.38	47.54	53.26	29
MGAFSSD	One-stage	50.65	43.09	39.65	44.46	56.09	48.64	44.90	49.82	10
HVPR	One-stage	53.47	43.96	45.96	40.46	-	-	-	-	36
SVGA-Net	One-stage	48.48	40.39	37.92	42.26	53.09	45.68	43.30	47.36	16
IA-SSD	One-stage	47.90	41.03	42.30	37.98	52.73	45.07	42.75	46.85	85
Proposed Method	**One-stage**	**55.74**	**45.28**	**47.27**	**42.05**	**61.24**	**52.47**	**49.41**	**54.37**	**30**